MAAPC Practicum Report

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1 Model and open loop analysis

1.1 (a)

The linear steady-state model is given by the system of equations

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx + Du. \end{cases}$$

The state variables are $x = [\theta, \alpha, \dot{\theta}, \dot{\alpha}]$,

where θ is the angle of the arm, measured from the reference point
α is the angle of the rod, from the vertical position
$\dot{\theta}$ is the angular velocity of the arm
$\dot{\alpha}$ is the angular velocity of the rod.

Numerical values of the model matrices:

$$A = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ 0 & 40.7056 & -13.2363 & 0 \\ 0 & 0.0760 & -4.7195 & 0 \end{bmatrix} \; B = \begin{bmatrix} 0 \\ 0 \\ 23.2996 \\ 8.3076 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \; D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

1.2 (b)

The poles of the system can be calculated using *Matlab* by determining the eigenvalues of the matrix A:

- -14.1907
- 0.4772 + 3.6386i
- 0.4772 - 3.6386i
- 0

The system does not have any transmission zeros, which is confirmed by the use of the function *tzero* returning an empty vector.

The controllability matrix of the system is given by $M = [B \quad AB \quad A^2B \quad A^3B]$. The rank of the matrix M is 4, which is equal to the number of states of the system. This means that the system is fully controllable, and therefore stabilizable.

The observability matrix of the system is given by $N = [C \quad CA \quad CA^2 \quad CA^3]^T$. The rank of the matrix N is 4, which is equal to the number of states of the system. This means that the system is fully observable, and therefore detectable.

Since the system is fully controllable and observable we can say that it is minimal.

1.3 (c)

The control goals for our system were mainly a fast response and good disturbance rejection, but after these were done, we were also able to implement setpoint tracking, allowing for the change of the system's equilibrium point. We strived for a settling time below 5 seconds.

2 Design of the LQR controller and creation of the first closed-loop simulation diagram

2.1 (a)

The diagonal values of the Q-matrix (Weighting matrix for states) and the R-matrix (Weighting matrix for control inputs) play a crucial role in the behaviour of the closed loop system. Increasing the diagonal values of the Q matrix leads to a higher penalization for deviations in the respective state (if you change the first value it affects the penalization of changes in θ and so on). This means that every deviation from this values equilibrium point will be met with a control action that penalizes it, and this penalization is proportional to the diagonal value chosen. The same occurs for the R-matrix but this time increasing the values leads to a smoother and less aggressive control action, while decreasing this value has the opposite effect.

This means that large Q values and small R values lead to aggressive control actions and faster response times and vice-versa.

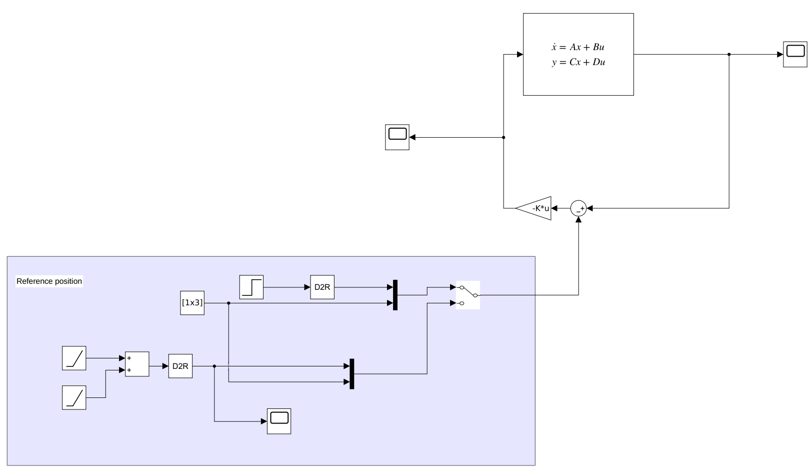

Figure 1: The Simulink diagram of linear system control with full state observations.

2.2 (b)

We started the design process with the following matrix:

$$Q = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 20 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad R = \begin{bmatrix} 1.5 \end{bmatrix}$$

After testing the step response we decided to penalize the changes in θ a bit more in order to get a faster response and a smaller settling time. We also decided to increase the value of R to get smoother movements and increase stability.

$$Q = \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 20 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad R = \begin{bmatrix} 2 \end{bmatrix}$$

Afterwards, we decided to also penalize $\dot{\theta}$ and $\dot{\alpha}$ to reduce the oscillations these presented around the equilibrium point and further increase the stability of the system. Finally we arrived at the following matrices which showed the best results:

$$Q = \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 20 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}, \qquad R = \begin{bmatrix} 2 \end{bmatrix}$$

2.3 (c)

As we can see in the plot below the system was disturbed with a change in setpoint from $\theta = 0$ to $\theta = 45$ which is about $0.785 radians$. The system reacts almost immediately and settles at the new setpoint in less than 5 seconds, as we set out to do.

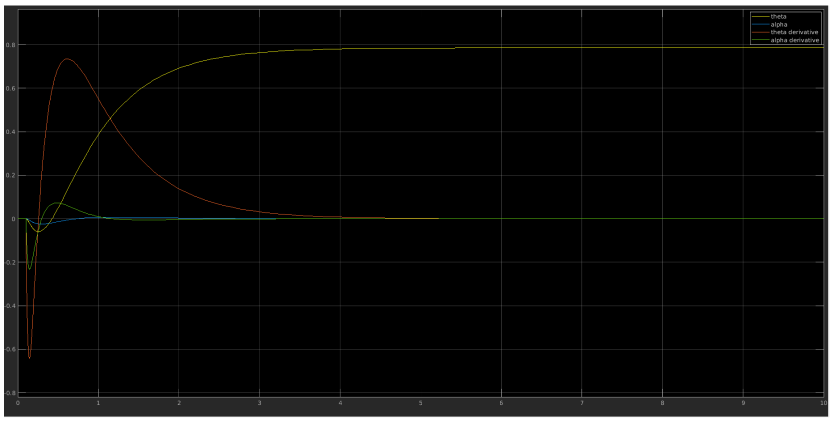

Figure 2: States of the system after reference θ changed to $45°$

The control actions taken by the system can be seen below.

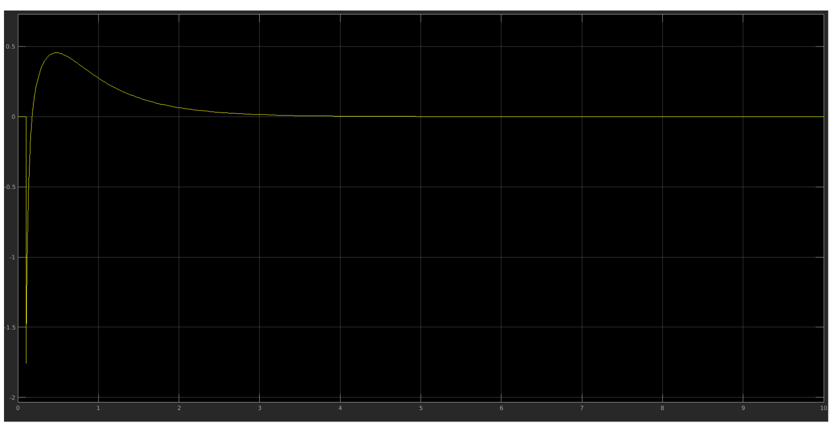

Figure 3: Control actions after reference θ changed to $45°$

3 Creation of the second closed-loop simulation diagram

3.1 (a)

The following Simulink diagram represents the second closed-loop simulation diagram.

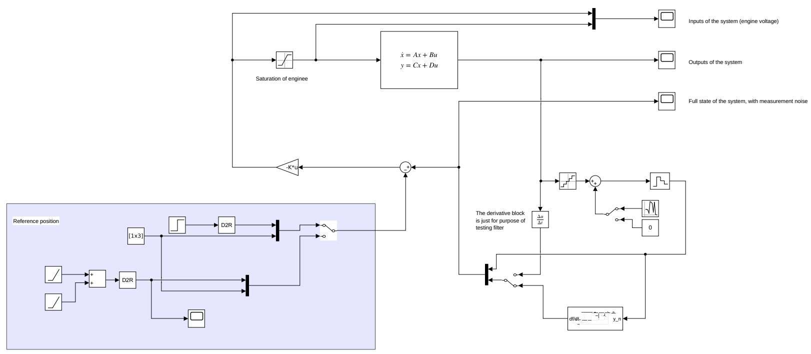

Figure 4: The Simulink diagram of linear system control with noise and quantization of measurements.

The purple block is used to set the setpoint of the system and provides two options: A step change in the θ setpoint of 45 that are then converted to radians and joined to an vector of 0s that is then passed to the input of the controller; The other option is a ramp that goes from 0s to 5s with a slope of 5 and stays constant after 5 seconds. This was used to test a continuous change in setpoint.

The saturation block guarantees that the input value to the system does not exceed the physical limits of the motor.

The gain block performs the control action.

The state-space block represents the model of the system.

The blocks on the bottom right of the diagram simulate the quantization effects of the A/D converters and the measurement noise from the sensors, as well as computing the derivatives of the measured outputs by using the backward difference equation and the low pass filter. The derivative block is only present for testing purposes.

The subsystem for the derivative estimation can be seen below:

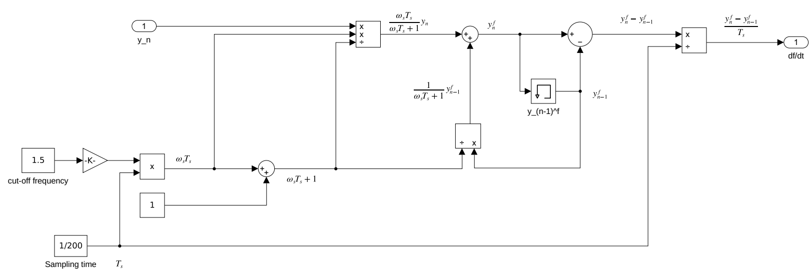

Figure 5: Derivative estimation with low pass filter.

The saturation limits were set in accordance to the limitations of the physical system.

The Quantization interval was set as 5ms in order to mimic the frequency at which the physical system can take measurements.

The measurement noise was set by looking at the measurements and trying to find a reasonable approximation. This amplitude could maybe be measured with realm measurements by leaving the rod still without any input and at constant θ and α angles and measuring the deviation of each measurement from the mean value.

3.2 (b)

The Q and R values set in previous stages performed well enough to where we could keep them the same. The new closed loop system benefits from the filter by presenting low levels of noise, but when compared with the system with the full state vector available, it does present some differences. This is to be expected as the derivative action is not fully reliable and the noise and quantization added also lead to a less accurate result.

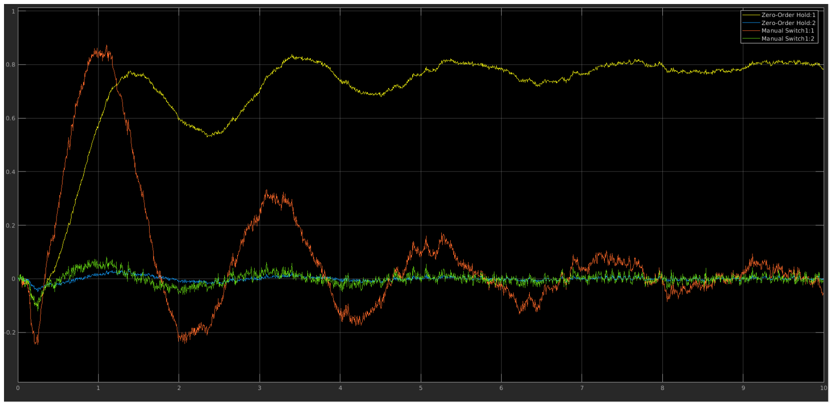

Figure 6: The response to 45 deg step with 0.5hz low pass filter.

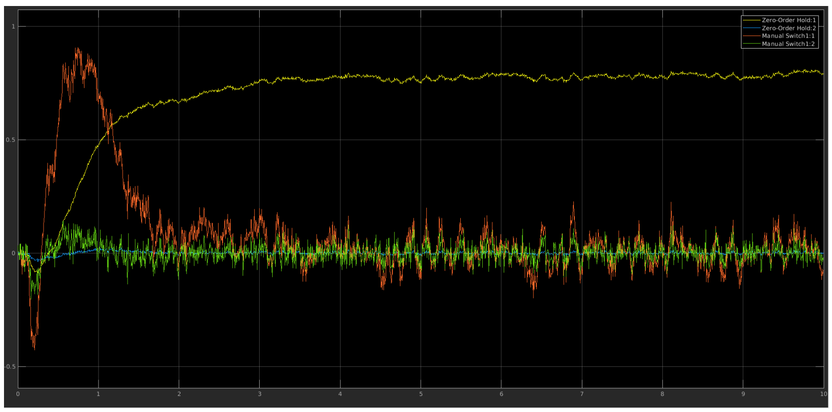

Figure 7: The response to 45 deg step with 1.5hz low pass filter.

3.3 (c)

The cutoff frequency plays an important role in reducing the effect of the measurement noise in the computation of the derivatives of the outputs and preserving the system's dynamics.

Setting it too low can lead to a slower response to changes and to some dynamics of the system being ignored. It can also, in extreme cases, lead to a phase lag which will affect the stability of the system.

Setting the cutoff frequency as a really high value can lead to some noise and other high-frequency signals going through the filter and leading to inaccurate derivative estimates. It can also result in jittery behavior and affect the stability of the system.

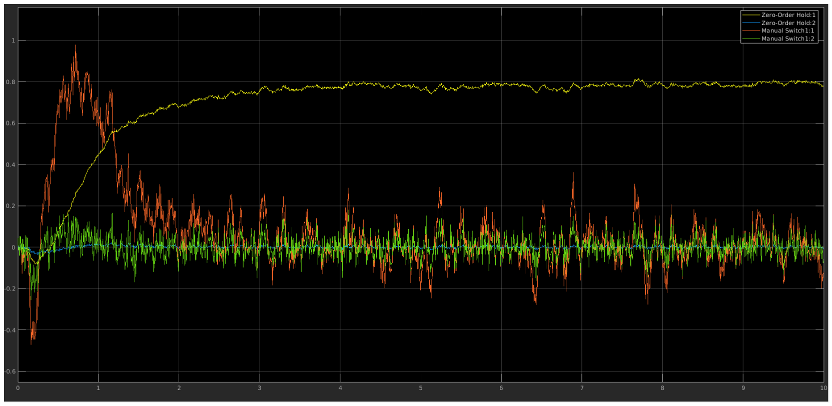

Figure 8: The response to 45 deg step with 2hz low pass filter.

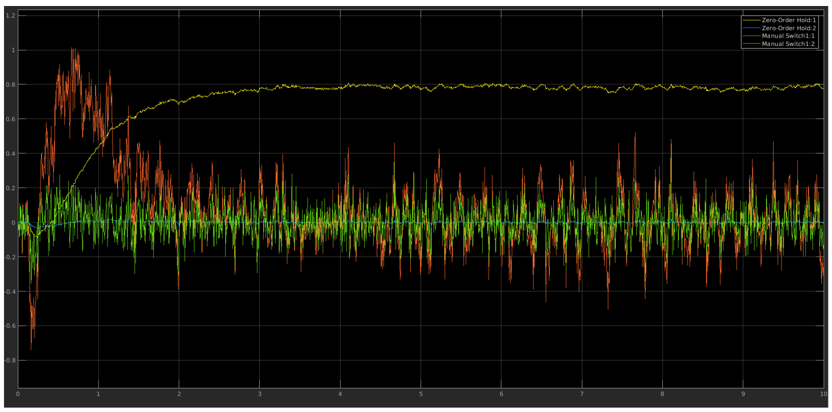

Figure 9: The response to 45 deg step with 4hz low pass filter.

4 Experimental results ("The proof of the pudding")

4.1 (a)

The derivative estimation subsystem is identical to the one created in the Section 3 and its diagram can be found at the Figure 5. It computes the derivatives of the outputs and applies the low-pass filter. The outputs and their derivations are then compared to the setpoints and fed to the controller, which then feeds the control action to the input of the system. There is a slider that allows us to change the setpoint of the system. We also added some displays so we could see the values of different variables in real time.

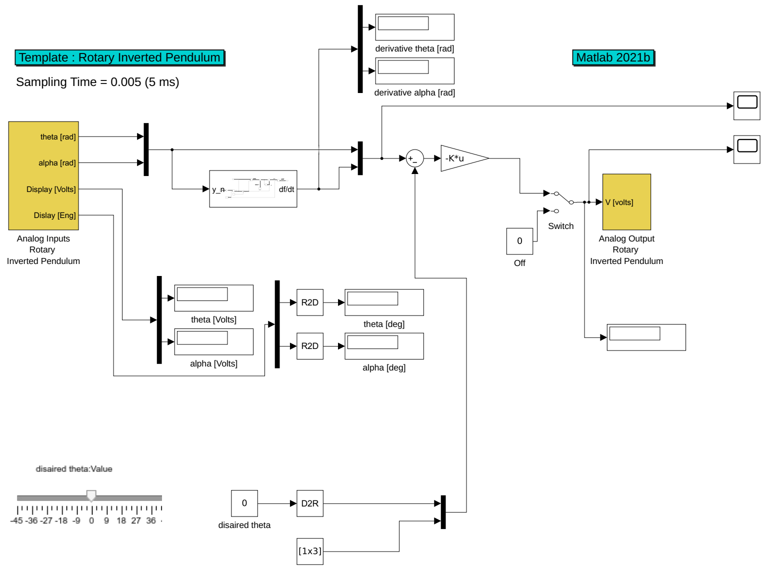

Figure 10: The Simulink diagram of the real-time system control.

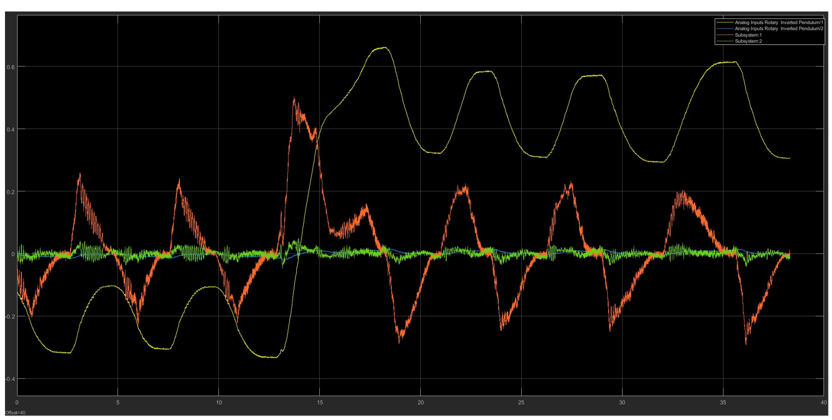

Figure 11: System response to the change of the reference by 45 degrees (first try).

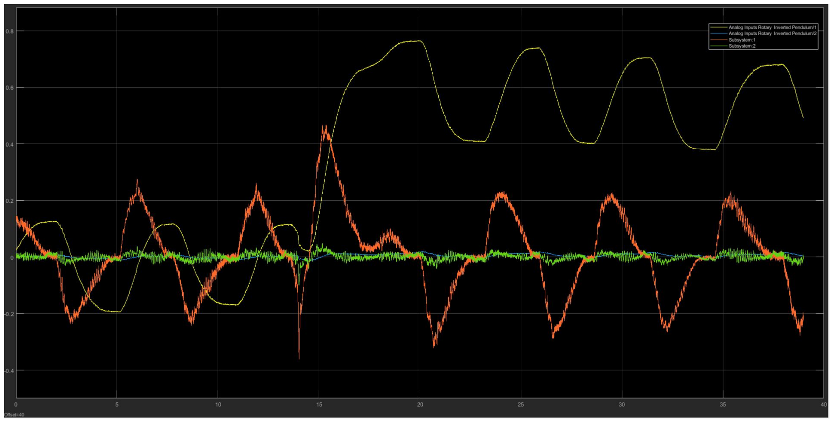

Figure 12: System response to the change of the reference by 45 degrees (second try).

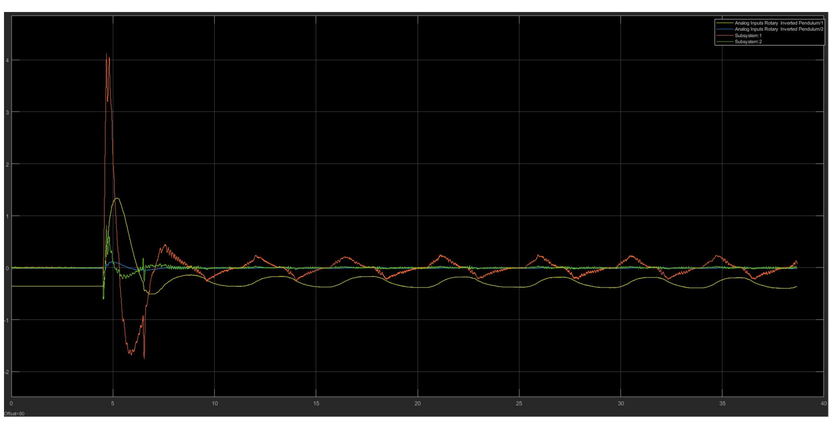

Figure 13: System response to the external disturbance (first try).

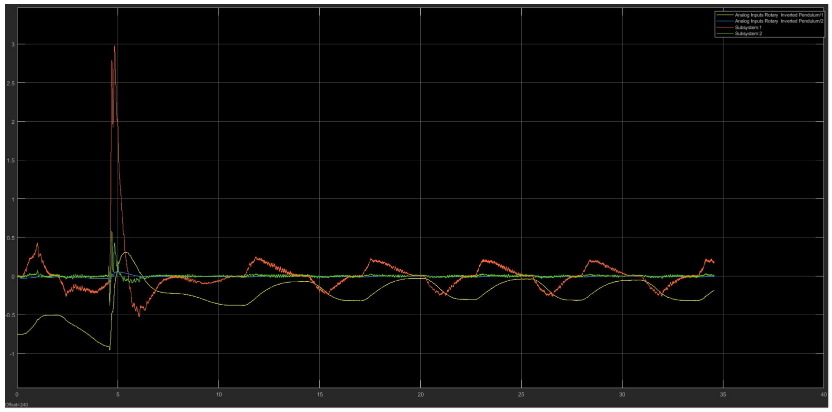

Figure 14: System response to the external disturbance (second try).

5 Conclusions

In this report, we have investigated the design and implementation of a controller for a rotary inverted pendulum system using the LQR approach. By analysing the system's nonlinear model and its linearization, we derived a state-space model that served as the basis for controller design.

The LQR controller was designed to optimize the closed-loop system's performance by minimizing the loss function. The weight matrices, Q and R, were designed to assign relative importance to minimizing the states and the input signal's influence. The MATLAB command *lqr* was utilized to calculate the optimal state-feedback gain, K.

The implementation of the controller was carried out in Simulink on a digital computer with a data acquisition board. The inputs of the system were obtained from measurements of the arm and rod angles and converted from voltage to engineering units and sampled at a frequency of 200 Hz. The derivative of the measured angles was estimated using a low-pass filter with a cutoff frequency.

We discovered that the cutoff frequency of the low-pass filter had a significant impact on the system's performance. A proper choice of cutoff frequency allowed for effective noise filtering while preserving the relevant system dynamics.

In conclusion, the successful design and implementation of the LQR controller demonstrated its effectiveness in regulating the rotary inverted pendulum system. The choice of weight matrices and cutoff frequency played vital roles in achieving desirable control performance. This project provided valuable insights into control system design, emphasizing the importance of parameter selection and the trade-offs involved in achieving optimal performance.